Mohd Qasim

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GitHub | LinkedIn | Portfolio

Academic Qualifications

Indian Institute of Technology, Jodhpur — Sep 2023 – May 2027
Bachelor of Technology in Chemical Engineering — *Jodhpur, Rajasthan*

Jawahar Navodaya Vidyalaya (JNV), Budgam — 2022
Class XII, CBSE — *Budgam, Jammu & Kashmir*

Mutahhary Public High School, Saliskote — 2020
Class X, JKBOSE — *Kargil, Ladakh*

Key Projects

- **AI for a Reactor-Separator Process**
  Mentor: Dr. Santosh Kumar Varanasi, Course: Application of AI in Chemical Engineering.
  - Studied a plant with two stirred-tank reactors (CSTRs) and a flash separator with vapour recycle and purge, modelling it from first principles with mass and energy balances that matched the published steady state to 4 decimal places.
  - Trained a neural network on 2,000 simulated steady states to learn how heat inputs Q1, Q2 and Q3 affect temperatures and compositions in all three vessels, reaching R^2 of 0.988 on held-out data.
  - Used a genetic algorithm to find operating settings that used **11.6% less heat** for nearly the same output.
  - Built an AI-based controller (fuzzy models, MPC) that tracked setpoint changes with less error than a tuned PI loop; also identified its limits, as the PI loop handled a sudden feed-temperature drop better and the learned dynamic model drifts over long prediction horizons.
  - Tools: Python, SciPy, scikit-learn, fuzzy models, genetic algorithm, MPC.
- **Catalytic Synthesis of 4-Methoxypropiophenone**
  Mentor: Prof. Sumit Kamal, Department of Chemical Engineering.
  - Helped synthesize 4-methoxypropiophenone by Friedel-Crafts acylation of anisole with propionic anhydride, using Amberlyst-15 as a solid acid catalyst (heterogeneous catalysis).
  - Ran the reaction in a temperature-controlled four-neck glass batch reactor with steady stirring.
  - Took samples over time to track product formation at different catalyst loadings.
- **Zeolite from Rice Husk for Clean Water**
  Mentor: Prof. Sumit Kamal, Department of Chemical Engineering.
  - Synthesized zeolite from rice husk, a low-cost silica source, through a hydrothermal process.
  - Characterized the material using XRD, SEM and FTIR.
  - Tested its adsorption performance for removing fluoride from drinking water.

Technical Skills

- **Programming Languages:** Python
- **Software and Libraries:** Aspen Plus, SciPy, Scikit-Learn
- **Process Modeling and AI Methods:** First-principles modeling, Neural networks, Fuzzy models, Genetic algorithm, Model Predictive Control (MPC)
- **Laboratory and Characterization:** Batch reactors, Heterogeneous catalysis, Adsorption, XRD, SEM, FTIR

Relevant Coursework

- **Computer Science:** Introduction to Computer Science, Introduction to Machine Learning
- **Mathematics:** Multivariable and Vector Calculus, Linear Algebra and Differential Equations, Probability, Statistics and Stochastic Processes
- **Core Engineering and Sciences:** Fluid Mechanics, Thermodynamics, Mass Transfer, Material and Energy Balance, Chemical Reaction Engineering, Molecular Engineering, Quantum Chemistry and Spectroscopy